Pandas

In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
data = {
    "StudentID": [101, 102, 103, 104, 105,
                  106, 107, 108, 109, 110,
                  111, 112, 113, 114, 115],

    "Name": ["Arjun", "Priya", "Rahul", "Sneha", "Kiran",
             "Divya", "Rohit", "Ananya", "Vikram", "Meena",
             "Suresh", "Kavya", "Nikhil", "Pooja", "Rahul"],

    "Department": ["CSE", "ECE", "CSE", "IT", "ECE",
                   "CSE", "IT", "CSE", "ECE", "IT",
                   "CSE", "ECE", "IT", "CSE", "CSE"],

    "Age": [20, 21, 20, 22, 21,
            20, 23, 21, 22, 20,
            21, 22, 20, 21, 20],

    "Marks": [85, 92, 76, 88, None,
              95, 67, 81, 73, 89,
              91, None, 78, 84, 76],

    "Attendance": [90, 95, 78, 88, 82,
                   98, None, 85, 75, 92,
                   96, 80, 86, 91, 78],

    "StudyHours": [3.5, 5.0, 2.0, 4.0, 3.0,
                   6.0, 1.5, 3.5, 2.5, 4.5,
                   5.5, 2.0, None, 4.0, 2.0]
}

df = pd.DataFrame(data)

df
```

Out[2]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
1	102	Priya	ECE	21	92.0	95.0	5.0
2	103	Rahul	CSE	20	76.0	78.0	2.0
3	104	Sneha	IT	22	88.0	88.0	4.0
4	105	Kiran	ECE	21	NaN	82.0	3.0
5	106	Divya	CSE	20	95.0	98.0	6.0
6	107	Rohit	IT	23	67.0	NaN	1.5
7	108	Ananya	CSE	21	81.0	85.0	3.5
8	109	Vikram	ECE	22	73.0	75.0	2.5
9	110	Meena	IT	20	89.0	92.0	4.5
10	111	Suresh	CSE	21	91.0	96.0	5.5
11	112	Kavya	ECE	22	NaN	80.0	2.0
12	113	Nikhil	IT	20	78.0	86.0	NaN
13	114	Pooja	CSE	21	84.0	91.0	4.0
14	115	Rahul	CSE	20	76.0	78.0	2.0

In [3]:
```
df.to_csv("students.csv", index=False)
```

In [4]:
```
df = pd.read_csv("students.csv")
```

In [5]:
```
df.head()
```

Out[5]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
1	102	Priya	ECE	21	92.0	95.0	5.0
2	103	Rahul	CSE	20	76.0	78.0	2.0
3	104	Sneha	IT	22	88.0	88.0	4.0
4	105	Kiran	ECE	21	NaN	82.0	3.0

In [6]:
```
df.head(3)
```

Out[6]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
1	102	Priya	ECE	21	92.0	95.0	5.0
2	103	Rahul	CSE	20	76.0	78.0	2.0

In [7]:
```
df.tail()
```

Out[7]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
10	111	Suresh	CSE	21	91.0	96.0	5.5
11	112	Kavya	ECE	22	NaN	80.0	2.0
12	113	Nikhil	IT	20	78.0	86.0	NaN
13	114	Pooja	CSE	21	84.0	91.0	4.0
14	115	Rahul	CSE	20	76.0	78.0	2.0

In [8]:
```
df.tail(2)
```

Out[8]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
13	114	Pooja	CSE	21	84.0	91.0	4.0
14	115	Rahul	CSE	20	76.0	78.0	2.0

In [10]:
```
df.shape
```

Out[10]:
```
(15, 7)
```

In [11]:
```
df.ndim
```

Out[11]:
```
2
```

In [12]:
```
df.columns
```

Out[12]:
```
Index(['StudentID', 'Name', 'Department', 'Age', 'Marks', 'Attendance',
       'StudyHours'],
      dtype='str')
```

In [13]:
```
df.dtypes
```

Out[13]:
```
StudentID       int64
Name              str
Department        str
Age             int64
Marks         float64
Attendance    float64
StudyHours    float64
dtype: object
```

In [14]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   StudentID   15 non-null     int64
 1   Name        15 non-null     str
 2   Department  15 non-null     str
 3   Age         15 non-null     int64
 4   Marks       13 non-null     float64
 5   Attendance  14 non-null     float64
 6   StudyHours  14 non-null     float64
dtypes: float64(3), int64(2), str(2)
memory usage: 972.0 bytes
```

In [15]:
```
df.describe()
```

Out[15]:

	StudentID	Age	Marks	Attendance	StudyHours
count	15.000000	15.000000	13.000000	14.000000	14.000000
mean	108.000000	20.933333	82.692308	86.714286	3.500000
std	4.472136	0.961150	8.350495	7.342485	1.414214
min	101.000000	20.000000	67.000000	75.000000	1.500000
25%	104.500000	20.000000	76.000000	80.500000	2.125000
50%	108.000000	21.000000	84.000000	87.000000	3.500000
75%	111.500000	21.500000	89.000000	91.750000	4.375000
max	115.000000	23.000000	95.000000	98.000000	6.000000

In [16]:
```
df.describe(include='all')
```

Out[16]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
count	15.000000	15	15	15.000000	13.000000	14.000000	14.000000
unique	NaN	14	3	NaN	NaN	NaN	NaN
top	NaN	Rahul	CSE	NaN	NaN	NaN	NaN
freq	NaN	2	7	NaN	NaN	NaN	NaN
mean	108.000000	NaN	NaN	20.933333	82.692308	86.714286	3.500000
std	4.472136	NaN	NaN	0.961150	8.350495	7.342485	1.414214
min	101.000000	NaN	NaN	20.000000	67.000000	75.000000	1.500000
25%	104.500000	NaN	NaN	20.000000	76.000000	80.500000	2.125000
50%	108.000000	NaN	NaN	21.000000	84.000000	87.000000	3.500000
75%	111.500000	NaN	NaN	21.500000	89.000000	91.750000	4.375000
max	115.000000	NaN	NaN	23.000000	95.000000	98.000000	6.000000

In [17]:
```
df.isnull()
```

Out[17]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	True	False	False
5	False	False	False	False	False	False	False
6	False	False	False	False	False	True	False
7	False	False	False	False	False	False	False
8	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False
10	False	False	False	False	False	False	False
11	False	False	False	False	True	False	False
12	False	False	False	False	False	False	True
13	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False

In [18]:
```
df.isna()
```

Out[18]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	True	False	False
5	False	False	False	False	False	False	False
6	False	False	False	False	False	True	False
7	False	False	False	False	False	False	False
8	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False
10	False	False	False	False	False	False	False
11	False	False	False	False	True	False	False
12	False	False	False	False	False	False	True
13	False	False	False	False	False	False	False
14	False	False	False	False	False	False	False

In [19]:
```
df.isnull().sum()
```

Out[19]:
```
StudentID     0
Name          0
Department    0
Age           0
Marks         2
Attendance    1
StudyHours    1
dtype: int64
```

In [20]:
```
df.isna().sum()
```

Out[20]:
```
StudentID     0
Name          0
Department    0
Age           0
Marks         2
Attendance    1
StudyHours    1
dtype: int64
```

In [21]:
```
df.isna().sum().sum()
```

Out[21]:
```
np.int64(4)
```

In [22]:
```
df.count()---Counting Non Null values
```

Out[22]:
```
StudentID     15
Name          15
Department    15
Age           15
Marks         13
Attendance    14
StudyHours    14
dtype: int64
```

In [23]:
```
df["Marks"].count()
```

Out[23]:
```
np.int64(13)
```

In [24]:
```
df.nunique()
```

Out[24]:
```
StudentID     15
Name          14
Department     3
Age            4
Marks         12
Attendance    13
StudyHours    10
dtype: int64
```

In [25]:
```
df["Department"].unique()
```

Out[25]:
```
<StringArray>
['CSE', 'ECE', 'IT']
Length: 3, dtype: str
```

In [26]:
```
df["Department"].nunique()
```

Out[26]:
```
3
```

In [27]:
```
df.duplicated()
```

Out[27]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
dtype: bool
```

In [28]:
```
df.duplicated().sum()
```

Out[28]:
```
np.int64(0)
```

In [29]:
```
df[df.duplicated()]
```

Out[29]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours

In [30]:
```
df["Name"].duplicated().sum()
```

Out[30]:
```
np.int64(1)
```

In [34]:
```
totmarks=df["Marks"].sum()
avgMarks=df["Marks"].mean()
maxMarks=df["Marks"].max()
minMarks=df["Marks"].min()

print("Total Marks :",totmarks)
print("Average Marks :",avgMarks)
print("Maximum Marks :",maxMarks)
print("Minimum Marks :",minMarks)
```

```
Total Marks : 1075.0
Average Marks : 82.6923076923077
Maximum Marks : 95.0
Minimum Marks : 67.0
```

In [35]:
```
df[df["Marks"] > 80]
```

Out[35]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
1	102	Priya	ECE	21	92.0	95.0	5.0
3	104	Sneha	IT	22	88.0	88.0	4.0
5	106	Divya	CSE	20	95.0	98.0	6.0
7	108	Ananya	CSE	21	81.0	85.0	3.5
9	110	Meena	IT	20	89.0	92.0	4.5
10	111	Suresh	CSE	21	91.0	96.0	5.5
13	114	Pooja	CSE	21	84.0	91.0	4.0

In [36]:
```
df[(df["Marks"] > 80) & (df["Attendance"] > 85)]
```

Out[36]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
1	102	Priya	ECE	21	92.0	95.0	5.0
3	104	Sneha	IT	22	88.0	88.0	4.0
5	106	Divya	CSE	20	95.0	98.0	6.0
9	110	Meena	IT	20	89.0	92.0	4.5
10	111	Suresh	CSE	21	91.0	96.0	5.5
13	114	Pooja	CSE	21	84.0	91.0	4.0

In [37]:
```
df[df["Department"] == "CSE"]
```

Out[37]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
2	103	Rahul	CSE	20	76.0	78.0	2.0
5	106	Divya	CSE	20	95.0	98.0	6.0
7	108	Ananya	CSE	21	81.0	85.0	3.5
10	111	Suresh	CSE	21	91.0	96.0	5.5
13	114	Pooja	CSE	21	84.0	91.0	4.0
14	115	Rahul	CSE	20	76.0	78.0	2.0

In [38]:
```
df[df["Department"] != "CSE"]
```

Out[38]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
1	102	Priya	ECE	21	92.0	95.0	5.0
3	104	Sneha	IT	22	88.0	88.0	4.0
4	105	Kiran	ECE	21	NaN	82.0	3.0
6	107	Rohit	IT	23	67.0	NaN	1.5
8	109	Vikram	ECE	22	73.0	75.0	2.5
9	110	Meena	IT	20	89.0	92.0	4.5
11	112	Kavya	ECE	22	NaN	80.0	2.0
12	113	Nikhil	IT	20	78.0	86.0	NaN

In [39]:
```
df[df["Marks"].between(70, 90)]
```

Out[39]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours
0	101	Arjun	CSE	20	85.0	90.0	3.5
2	103	Rahul	CSE	20	76.0	78.0	2.0
3	104	Sneha	IT	22	88.0	88.0	4.0
7	108	Ananya	CSE	21	81.0	85.0	3.5
8	109	Vikram	ECE	22	73.0	75.0	2.5
9	110	Meena	IT	20	89.0	92.0	4.5
12	113	Nikhil	IT	20	78.0	86.0	NaN
13	114	Pooja	CSE	21	84.0	91.0	4.0
14	115	Rahul	CSE	20	76.0	78.0	2.0

In [40]:
```
df.groupby("Department")["Marks"].mean()
```

Out[40]:
```
Department
CSE    84.0
ECE    82.5
IT     80.5
Name: Marks, dtype: float64
```

In [41]:
```
df.groupby("Department")["Marks"].agg(
    ["count", "mean", "min", "max", "sum"]
)
```

Out[41]:

Department	count	mean	min	max	sum
CSE	7	84.0	76.0	95.0	588.0
ECE	2	82.5	73.0	92.0	165.0
IT	4	80.5	67.0	89.0	322.0

In [42]:
```
df.groupby(["Department", "Age"])["Marks"].mean()
```

Out[42]:
```
Department  Age
CSE         20     83.000000
            21     85.333333
ECE         21     92.000000
            22     73.000000
IT          20     83.500000
            22     88.000000
            23     67.000000
Name: Marks, dtype: float64
```

In [43]:
```
df.groupby("Department").size()
```

Out[43]:
```
Department
CSE    7
ECE    4
IT     4
dtype: int64
```

In [44]:
```
df["PerformanceScore"] = (
    df["Marks"] * 0.8 +
    df["Attendance"] * 0.2
)

df.head()
```

Out[44]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours	PerformanceScore
0	101	Arjun	CSE	20	85.0	90.0	3.5	86.0
1	102	Priya	ECE	21	92.0	95.0	5.0	92.6
2	103	Rahul	CSE	20	76.0	78.0	2.0	76.4
3	104	Sneha	IT	22	88.0	88.0	4.0	88.0
4	105	Kiran	ECE	21	NaN	82.0	3.0	NaN

In [45]:
```
df["PerformanceScore"].describe()
```

Out[45]:
```
count    12.000000
mean     84.733333
std       7.207740
min      73.400000
25%      78.800000
50%      85.700000
75%      90.200000
max      95.600000
Name: PerformanceScore, dtype: float64
```

In [47]:
```
avgPerformance=df["PerformanceScore"].mean()
medianPerformance=df["PerformanceScore"].median()
standardDeviation=df["PerformanceScore"].std()
maxPerformance=df["PerformanceScore"].max()
minPerformance=df["PerformanceScore"].min()
totPerformance=df["PerformanceScore"].sum()

print("Average Performance:",avgPerformance)
print("Median of Performance:",medianPerformance)
print("standardDeviation of Performance:",standardDeviation)
print("Maximun Performance:",maxPerformance)
print("Minimum Performance:",minPerformance)
print("Total Performance",totPerformance)
```

```
Average Performance: 84.73333333333333
Median of Performance: 85.7
standardDeviation of Performance: 7.207739947550489
Maximun Performance: 95.6
Minimum Performance: 73.4
Total Performance 1016.8
```

In [48]:
```
df[df["PerformanceScore"] > 80]
```

Out[48]:

	StudentID	Name	Department	Age	Marks	Attendance	StudyHours	PerformanceScore
0	101	Arjun	CSE	20	85.0	90.0	3.5	86.0
1	102	Priya	ECE	21	92.0	95.0	5.0	92.6
3	104	Sneha	IT	22	88.0	88.0	4.0	88.0
5	106	Divya	CSE	20	95.0	98.0	6.0	95.6
7	108	Ananya	CSE	21	81.0	85.0	3.5	81.8
9	110	Meena	IT	20	89.0	92.0	4.5	89.6
10	111	Suresh	CSE	21	91.0	96.0	5.5	92.0
13	114	Pooja	CSE	21	84.0	91.0	4.0	85.4

In [51]:
```
df[['StudentID', 'Name']]
```

Out[51]:

	StudentID	Name
0	101	Arjun
1	102	Priya
2	103	Rahul
3	104	Sneha
4	105	Kiran
5	106	Divya
6	107	Rohit
7	108	Ananya
8	109	Vikram
9	110	Meena
10	111	Suresh
11	112	Kavya
12	113	Nikhil
13	114	Pooja
14	115	Rahul

Numpy

In [52]:
```
import numpy as np
```

In [53]:
```
temperatures = np.array([28, 32, 35, 30, 27, 38, 33])

print(temperatures)
```

```
[28 32 35 30 27 38 33]
```

In [54]:
```
print("Dimensions:", temperatures.ndim)
print("Shape:", temperatures.shape)
print("Size:", temperatures.size)
```

```
Dimensions: 1
Shape: (7,)
Size: 7
```

In [55]:
```
print(temperatures + 2)
```

```
[30 34 37 32 29 40 35]
```

In [56]:
```
print(temperatures - 5)
```

```
[23 27 30 25 22 33 28]
```

In [57]:
```
print(temperatures * 2)
```

```
[56 64 70 60 54 76 66]
```

In [58]:
```
print(temperatures / 2)
```

```
[14.  16.  17.5 15.  13.5 19.  16.5]
```

In [59]:
```
morning_temp = np.array([22, 24, 25, 23, 21, 26, 24])
evening_temp = np.array([28, 32, 35, 30, 27, 38, 33])

print(evening_temp - morning_temp)
```

```
[ 6  8 10  7  6 12  9]
```

In [60]:
```
print("Total:", np.sum(temperatures))
print("Average:", np.mean(temperatures))
print("Maximum:", np.max(temperatures))
print("Minimum:", np.min(temperatures))
```

```
Total: 223
Average: 31.857142857142858
Maximum: 38
Minimum: 27
```

In [61]:
```
print(temperatures > 32)
```

```
[False False  True False False  True  True]
```

In [62]:
```
print(temperatures[temperatures > 32])
```

```
[35 38 33]
```

In [63]:
```
print(np.sum(temperatures > 32))
```

```
3
```

In [64]:
```
weather = np.array([
    [32, 65, 12],
    [28, 80, 8],
    [35, 55, 18],
    [30, 70, 10],
    [38, 45, 22]
])

print(weather)
```

```
[[32 65 12]
 [28 80  8]
 [35 55 18]
 [30 70 10]
 [38 45 22]]
```

In [65]:
```
print("Dimensions:", weather.ndim)
print("Shape:", weather.shape)
print("Size:", weather.size)
```

```
Dimensions: 2
Shape: (5, 3)
Size: 15
```

In [66]:
```
print(np.sum(weather))
```

```
548
```

In [67]:
```
print(np.sum(weather, axis=0))
```

```
[163 315  70]
```

In [68]:
```
print(np.sum(weather, axis=1))
```

```
[109 116 108 110 105]
```

In [69]:
```
print(np.mean(weather, axis=0))
```

```
[32.6 63.  14. ]
```

In [70]:
```
print(np.max(weather, axis=0))
```

```
[38 80 22]
```

In [71]:
```
print(np.min(weather, axis=0))
```

```
[28 45  8]
```

In [72]:
```
print(np.max(weather, axis=1))
```

```
[65 80 55 70 45]
```

In [73]:
```
print(weather + 5)
```

```
[[37 70 17]
 [33 85 13]
 [40 60 23]
 [35 75 15]
 [43 50 27]]
```

In [74]:
```
print(weather * 2)
```

```
[[ 64 130  24]
 [ 56 160  16]
 [ 70 110  36]
 [ 60 140  20]
 [ 76  90  44]]
```

In [75]:
```
print(weather > 30)
```

```
[[ True  True False]
 [False  True False]
 [ True  True False]
 [False  True False]
 [ True  True False]]
```

In [76]:
```
print(weather[weather > 30])
```

```
[32 65 80 35 55 70 38 45]
```

In [79]:
```
print(weather[0])
```

```
[32 65 12]
```

In [80]:
```
print(weather[:, 0])
```

```
[32 28 35 30 38]
```

In [81]:
```
print(weather[:, 1])
```

```
[65 80 55 70 45]
```

In [77]:
```
print(weather[weather[:, 0] > 30])
```

```
[[32 65 12]
 [35 55 18]
 [38 45 22]]
```

In [78]:
```
print(weather[(weather[:, 0] > 30) & (weather[:, 1] < 60)])
```

```
[[35 55 18]
 [38 45 22]]
```

In [82]:
```
print(np.sort(temperatures))
```

```
[27 28 30 32 33 35 38]
```

In [83]:
```
values = np.array([10, 20, 30, 40, 50, 60])

print(values.reshape(2, 3))
```

```
[[10 20 30]
 [40 50 60]]
```